

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416605
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Bivolarsky; Lazar et al.

Acoustic signal material identification with nanotube couplant

Abstract

A system and method for improved, non-intrusive material identification includes a vessel holding or transporting at least one quantity of a fluid. At least one acoustic transducer is positioned on an exterior surface of a sidewall of the vessel. At least one coupling layer is positioned between the at least one acoustic transducer and the exterior surface of the sidewall of the vessel, wherein the at least one coupling layer is formed, at least in part, from nanotubes, wherein the nanotubes improve a reflection of an acoustic signal emitted from the at least one acoustic transducer. A computerized device is in communication with the at least one acoustic signal. The computerized device has a processor and a memory, and determines a material identification of the quantity of fluid based on, at least in part, the reflection of the ultrasonic signal.

Inventors: Bivolarsky; Lazar (Cupertino, CA), Burcham; Joel D. (Huntsville, AL), Coleman; William (Tucson, AZ), Heim; James M. (Tucson, AZ)

Applicant: PERCEPTIVE SENSOR TECHNOLOGIES, INC. (Tucson, AZ)

Family ID: 1000008820333

Assignee: PERCEPTIVE SENSOR TECHNOLOGIES, INC. (Tucson, AZ)

Appl. No.: 18/616718

Filed: March 26, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250052725 A1	Feb. 13, 2025

Related U.S. Application Data

continuation parent-doc US 18223987 20230719 US 11940420 child-doc US 18616718
us-provisional-application US 63368856 20220719

Publication Classification

Int. Cl.: G01N29/28 (20060101); G01N29/02 (20060101)

U.S. Cl.:

CPC G01N29/28 (20130101); G01N29/02 (20130101); G01N2291/022 (20130101); G01N2291/106 (20130101)

Field of Classification Search

CPC: G01N (29/28)

USPC: 73/628

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2449054	12/1947	Cantlin	200/84R	G01F 23/0038
3019650	12/1961	Worswick	73/290V	G01F 23/2961
3703829	12/1971	Dougherty	73/290	N/A
3837218	12/1973	Flambard et al.	N/A	G01N 29/00
3971962	12/1975	Green	N/A	H01L 41/08
4065958	12/1977	Krylova et al.	N/A	G01N 29/02
4118983	12/1977	Braznikov	N/A	G01F 23/28
4121468	12/1977	Glover et al.	N/A	G01N 29/04
4182177	12/1979	Prough	73/290	N/A
4208908	12/1979	Hickox	N/A	G01F 1/66
4280126	12/1980	White	340/621	N/A
4320659	12/1981	Lynnworth et al.	N/A	G01N 29/02
4326173	12/1981	Newman	N/A	H03L 7/08
4501146	12/1984	Greenhalgh	N/A	G01F 23/28
4580448	12/1985	Skrgatic	N/A	G01F 23/28
4596266	12/1985	Kinghorn et al.	N/A	B65D 88/38
4599892	12/1985	Doshi	N/A	G01F 17/00
4676098	12/1986	Erlenkämper et al.	73/290	N/A

4852416	12/1988	Boone et al.	N/A	H04R 1/02
4934191	12/1989	Kroening et al.	73/592	N/A
4954997	12/1989	Dieulesaint et al.	N/A	G08B 21/00
4977780	12/1989	Machida et al.	N/A	G01N 29/04
5015995	12/1990	Holroyd	340/621	N/A
5038611	12/1990	Weldon et al.	73/290	N/A
5040415	12/1990	Barkhoudarian	N/A	G01F 1/66
5148700	12/1991	King	N/A	G01N 15/00
5195058	12/1992	Simon	N/A	G01S 15/02
5223822	12/1992	Stommes et al.	N/A	G08B 7/06
5295120	12/1993	McShane	367/188	N/A
5325727	12/1993	Miller et al.	N/A	G01F 1/34
5415033	12/1994	Maresca, Jr. et al.	73/40.5	N/A
5438868	12/1994	Holden et al.	73/290	N/A
5453944	12/1994	Baumoel	N/A	G06F 17/00
5460046	12/1994	Maltby et al.	N/A	G01N 29/24
5469749	12/1994	Shimada et al.	N/A	G01F 1/38
5604314	12/1996	Grahn	N/A	G01L 5/16
5663505	12/1996	Nakamura	N/A	G01L 9/0022
5770806	12/1997	Hiismaki	N/A	G01F 1/662
5821427	12/1997	Byrd	N/A	G01F 1/66
5836192	12/1997	Getman et al.	N/A	G01F 23/28
6035903	12/1999	Few et al.	N/A	B65B 1/04
6105431	12/1999	Duffill et al.	N/A	G01N 29/2487
6151956	12/1999	Takahashi et al.	N/A	G01N 3/56
6157894	12/1999	Hess et al.	702/54	N/A
6192751	12/2000	Stein et al.	N/A	G08B 21/00
6330831	12/2000	Lynnworth et al.	73/861.28	N/A
6354147	12/2001	Gysling et al.	N/A	G01N 29/00
6368281	12/2001	Solomon et al.	N/A	A61B 8/14
6443006	12/2001	Degrave	N/A	G02F 23/00
6470744	12/2001	Usui et al.	73/290	N/A
6481287	12/2001	Ashworth et al.	N/A	G01K 11/24
6513385	12/2002	Han	N/A	G01N 29/00
6575043	12/2002	Huang et al.	N/A	G01F 1/66

6578424	12/2002	Ziola et al.	N/A	G01N 29/00
6631639	12/2002	Dam et al.	73/290	N/A
6672163	12/2003	Han et al.	N/A	G01V 1/28
6691582	12/2003	Nawa et al.	N/A	G01F 1/66
6836734	12/2003	Rojas et al.	N/A	G06F 19/00
6925868	12/2004	Young et al.	73/290	N/A
6938488	12/2004	Diaz et al.	73/597	N/A
7085391	12/2005	Yamaya	N/A	G10H 1/00
7114375	12/2005	Panetta et al.	73/61.75	N/A
7246522	12/2006	Diaz et al.	73/597	N/A
7299136	12/2006	DiFoggio et al.	702/22	N/A
7330797	12/2007	Bailey et al.	N/A	G01F 23/00
7359803	12/2007	Gysling et al.	N/A	G01N 31/00
7363174	12/2007	Kishiro et al.	N/A	G01F 25/10
7430924	12/2007	Gysling et al.	N/A	G01F 1/00
7437946	12/2007	Gysling et al.	N/A	G01F 1/22
7624650	12/2008	Gysling et al.	N/A	G01F 1/66
7624651	12/2008	Fernald et al.	N/A	G01F 1/66
7656747	12/2009	Mandal et al.	N/A	G01V 1/44
7694570	12/2009	Dam et al.	73/644	N/A
7757560	12/2009	Hofmann	N/A	G01R 33/20
7962293	12/2010	Gysling	N/A	G01F 1/76
7966882	12/2010	Greenwood	73/597	N/A
8249829	12/2011	Vass et al.	N/A	G06F 19/00
8346491	12/2012	Loose et al.	N/A	G01F 1/00
8482295	12/2012	Sadri et al.	N/A	G01R 27/04
8683882	12/2013	Jackson	N/A	G01N 9/24
8820182	12/2013	Nikolay Nikolov et al.	N/A	H04Q 9/00
8850882	12/2013	Qu et al.	N/A	G01F 23/296
8915145	12/2013	Van Orsdol	N/A	G01F 1/74
9057677	12/2014	Field	N/A	G01N 29/032
9383476	12/2015	Trehan et al.	N/A	G01V 11/00
9557208	12/2016	Kuroda et al.	N/A	G01F 23/28
9772311	12/2016	Liljenberg et al.	N/A	G01N 29/032
9816848	12/2016	Raykhman et al.	N/A	G01F 1/86

9835450	12/2016	Deleye et al.	N/A	G01N 29/069
9891085	12/2017	Muhammad et al.	N/A	G01F 1/88
9903840	12/2017	Altpeter et al.	N/A	G01N 29/04
10122051	12/2017	Kuhne et al.	N/A	H01M 10/484
10180410	12/2018	Takahashi et al.	N/A	G01N 29/043
10215613	12/2018	Kassubek et al.	N/A	G01F 23/296
10458871	12/2018	Norli	N/A	G01L 11/04
10794871	12/2019	Blackshire et al.	N/A	G01N 29/265
11020793	12/2020	De Monte et al.	N/A	B22D 2/006
11047721	12/2020	Schöb et al.	N/A	G01F 1/66
11274952	12/2021	Bober et al.	N/A	G01F 1/66
11293791	12/2021	Firouzi et al.	N/A	G01F 1/66
11536696	12/2021	Bivolarsky et al.	N/A	G01N 29/44
11585690	12/2022	Bivolarsky et al.	N/A	G01F 23/2965
11729537	12/2022	Heim et al.	N/A	G01N 29/221
11788904	12/2022	Bivolarsky et al.	N/A	G01K 3/14
2002/0170753	12/2001	Clare	N/A	G01G 19/22
2002/0173230	12/2001	Mayes	N/A	B24B 49/00
2004/0035208	12/2003	Diaz et al.	N/A	G01N 29/18
2004/0079150	12/2003	Breed et al.	73/291	N/A
2004/0173021	12/2003	Lizon et al.	73/290	N/A
2004/0226615	12/2003	Morikawa et al.	N/A	G05D 7/06
2005/0055136	12/2004	Hofmann et al.	700/273	N/A
2005/0102109	12/2004	Dubois et al.	N/A	G01B 5/28
2005/0128873	12/2004	LaBry	N/A	G01V 1/40
2005/0178198	12/2004	Freger et al.	73/290	N/A
2005/0247070	12/2004	Arshansky et al.	62/77	N/A
2006/0196224	12/2005	Esslinger	62/509	N/A
2007/0001028	12/2006	Gysling	N/A	B05B 7/30
2007/0068248	12/2006	Freger et al.	N/A	G01F 23/28
2007/0068253	12/2006	Carodiskey	N/A	A61B 6/00
2007/0157737	12/2006	Gysling et al.	N/A	G01F 1/667
2007/0205907	12/2006	Schenk, Jr.	N/A	G08B 21/00

2008/0092623	12/2007	Lynch et al.	N/A	G01N 29/02
2008/0101158	12/2007	Hosseini et al.	N/A	G01S 15/00
2009/0007678	12/2008	Fukutomi et al.	N/A	G02N 29/2487
2009/0143681	12/2008	Jurvelin et al.	N/A	A61B 8/00
2010/0046576	12/2009	Desai	N/A	G01K 11/22
2010/0111133	12/2009	Yuhas et al.	N/A	G01K 17/00
2010/0199779	12/2009	Liu et al.	N/A	G01F 1/663
2010/0218599	12/2009	Young et al.	N/A	G01F 23/296
2010/0242593	12/2009	Lagergren et al.	N/A	G01F 23/296
2010/0307249	12/2009	Lesage et al.	N/A	G01N 29/04
2011/0029262	12/2010	Barkhouse	702/55	N/A
2011/0048551	12/2010	Tanaka et al.	137/486	N/A
2011/0072904	12/2010	Lam et al.	N/A	G01N 29/04
2011/0120218	12/2010	Aldridge	73/290	N/A
2011/0239769	12/2010	Schmitt et al.	N/A	G01N 29/02
2011/0271769	12/2010	Kippersund et al.	N/A	G01F 1/66
2011/0284288	12/2010	Sawyer et al.	N/A	E21B 49/005
2012/0024067	12/2011	Oberdoerfer et al.	N/A	G01N 29/00
2012/0055239	12/2011	Sinha	N/A	G01N 29/00
2012/0173169	12/2011	Skelding	N/A	G06F 19/00
2012/0222471	12/2011	Raykhman et al.	N/A	G01N 29/02
2012/0226159	12/2011	Sinclair et al.	N/A	G01S 7/52046
2012/0259560	12/2011	Woltring et al.	702/55	N/A
2012/0262472	12/2011	Garr et al.	N/A	G06T 11/206
2012/0265454	12/2011	Rudd et al.	N/A	G01F 1/66
2012/0281096	12/2011	Gellaboina et al.	N/A	G01S 15/89
2013/0002443	12/2012	Breed et al.	N/A	G08B 21/00
2013/0041597	12/2012	Deleye et al.	N/A	G01N 29/043

2013/0068027	12/2012	Sullivan et al.	N/A	G01N 29/04
2013/0080081	12/2012	Dugger et al.	N/A	G01F 1/663
2013/0090575	12/2012	Rupp et al.	N/A	A61N 7/00
2013/0120155	12/2012	Hagg	N/A	G08C 19/16
2013/0128035	12/2012	Johns et al.	348/135	N/A
2013/0213714	12/2012	Fuida	N/A	E21B 49/00
2014/0020478	12/2013	Ao et al.	N/A	G01F 1/66
2014/0027455	12/2013	Castellano et al.	N/A	B65D 88/34
2014/0076415	12/2013	Dunki-Jacobs et al.	N/A	E03C 1/02
2014/0107435	12/2013	Sharf et al.	N/A	A61B 8/00
2014/0223992	12/2013	Harper et al.	N/A	G01F 25/0084
2014/0301902	12/2013	Fernald et al.	N/A	B01J 19/10
2014/0375169	12/2013	Na et al.	N/A	H01L 41/08
2015/0068311	12/2014	Tanaka et al.	N/A	G01N 17/00
2015/0075278	12/2014	Dockendorff et al.	N/A	G01F 23/296
2015/0177045	12/2014	Cobianu et al.	N/A	G01F 23/296
2015/0198471	12/2014	Furlong et al.	N/A	G01F 1/66
2015/0212045	12/2014	Raykhman et al.	N/A	G01F 1/74
2015/0247751	12/2014	Kutlik et al.	N/A	G01F 23/2962
2015/0260003	12/2014	McHugh et al.	N/A	E21B 33/076
2015/0276463	12/2014	Milne et al.	N/A	G01F 23/296
2015/0369647	12/2014	Kumar et al.	N/A	G01F 23/284
2016/0025545	12/2015	Saltzgiver et al.	N/A	G01F 23/263
2016/0041024	12/2015	Reimer et al.	N/A	G01F 23/296
2016/0108730	12/2015	Fanini et al.	N/A	E21B 49/08
2016/0109304	12/2015	Yan et al.	N/A	G01K 15/00
2016/0146653	12/2015	Skelding	N/A	E21B 21/01
2016/0169839	12/2015	Gottlieb et al.	N/A	G01N 29/02
2016/0216141	12/2015	Leaders et al.	N/A	G01F 1/66

2016/0265954	12/2015	Bachmann et al.	N/A	G01F 1/667
2016/0320226	12/2015	Schaefer et al.	N/A	G01F 23/296
2017/0002954	12/2016	Brown et al.	N/A	F16K 37/0058
2017/0010144	12/2016	Lenner et al.	N/A	G01F 23/296
2017/0010145	12/2016	Lenner et al.	N/A	G01F 23/2962
2017/0010146	12/2016	Kassubek et al.	N/A	G01F 23/2962
2017/0059389	12/2016	Moore et al.	N/A	G01F 23/2968
2017/0082650	12/2016	Hies et al.	N/A	G01F 25/0007
2017/0087526	12/2016	Luharuka	N/A	B01F 15/00
2017/0097322	12/2016	Giese et al.	N/A	G01N 29/07
2017/0102095	12/2016	Kunita et al.	N/A	F16K 37/0091
2017/0199295	12/2016	Mandal	N/A	G01V 1/50
2017/0202595	12/2016	Shelton, IV	N/A	A61B 18/00
2017/0239741	12/2016	Furuta	N/A	B23H 1/10
2017/0268915	12/2016	Gestner et al.	N/A	G01F 1/66
2017/0295743	12/2016	Brown et al.	N/A	A01J 5/0133
2017/0309989	12/2016	Waelde et al.	N/A	H01Q 1/225
2018/0035603	12/2017	Kremmer et al.	N/A	A01C 7/20
2018/0044159	12/2017	Crouse et al.	N/A	B67D 1/0406
2018/0080809	12/2017	Tokarev et al.	N/A	G01F 23/2965
2018/0113663	12/2017	Jain	N/A	G06F 3/14
2018/0149505	12/2017	Ploss et al.	N/A	G01F 1/66
2018/0266874	12/2017	Montoya et al.	N/A	G01F 23/68
2018/0299317	12/2017	Truong et al.	N/A	G01F 23/2925
2018/0306628	12/2017	Parrott et al.	N/A	G01F 17/00
2018/0348169	12/2017	Lee et al.	N/A	G01N 29/11
2019/0011304	12/2018	Cunningham et al.	N/A	G01F 17/00
2019/0063984	12/2018	Bowley	N/A	G01F 23/2962

2019/0078927	12/2018	Takayama et al.	N/A	G01F 23/2965
2019/0137310	12/2018	Xiao et al.	N/A	G01F 1/06
2019/0154480	12/2018	Schöb et al.	N/A	G01F 1/662
2019/0195629	12/2018	Vaissiere	N/A	G01C 9/00
2019/0195830	12/2018	Tamura et al.	N/A	N/A
2019/0272496	12/2018	Moeller	N/A	G06Q 10/087
2019/0368908	12/2018	Aughton et al.	N/A	G01F 1/66
2020/0018628	12/2019	Head et al.	N/A	G21C 17/022
2020/0182736	12/2019	Kim et al.	N/A	G01M 3/2807
2020/0195449	12/2019	Obaidi	N/A	H04L 9/38
2020/0200711	12/2019	Fehran et al.	N/A	G01N 29/024
2020/0378283	12/2019	Zhang et al.	N/A	F01M 11/10
2020/0378812	12/2019	Heim	N/A	G01F 1/667
2020/0378818	12/2019	Heim et al.	N/A	G01F 23/296
2021/0382014	12/2020	Xu et al.	N/A	G01N 29/24
2022/0034850	12/2021	Zhang et al.	N/A	G01N 29/22
2022/0178879	12/2021	Bivolarsky et al.	N/A	G02N 29/028
2022/0178881	12/2021	Bivolarsky et al.	N/A	G01N 29/04
2022/0276102	12/2021	Bivolarsky et al.	N/A	G01K 11/24
2023/0258488	12/2022	Coleman et al.	N/A	G01F 1/66

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
204944617	12/2015	CN	G01F 23/296
105333925	12/2015	CN	G01F 23/296
105548370	12/2015	CN	G01N 29/24
10 2010 029 254	12/2010	DE	F01N 3/10
0372700	12/1989	EP	G01F 23/28
2450701	12/2011	EP	G01N 29/22
2962096	12/2018	EP	G01L 1/255
2192717	12/1989	GB	G01N 29/00
H1073385	12/1997	JP	F28D 15/02
2000314651	12/1999	JP	G01F 23/28
2002340654	12/2001	JP	G01F 23/28
2013140029	12/2012	JP	F01K 5/02

200174618	12/1999	KR	G01N 29/24
WO 87/04793	12/1986	WO	G01N 29/00
WO 8809895	12/1987	WO	F16K 37/00
WO 9010849	12/1989	WO	G01F 23/28
WO 2007/149605	12/2006	WO	N/A
WO 2008/079202	12/2007	WO	G01F 3/36
WO 2009/154719	12/2008	WO	G01F 17/00
WO 2014/021846	12/2013	WO	G01F 1/66
WO 2014/167471	12/2013	WO	G01F 23/30
WO 2020/136945	12/2019	WO	G01F 23/296

OTHER PUBLICATIONS

Amjad, Umar et al, "Advanced signal processing technique for damage detection in steel tubes" Proceedings of SPIE, Health Monitoring of Structural and Biological Systems 2016, 980511 (Apr. 1, 2016);14 pgs. cited by applicant

Amjad, Umar et al. "Change in time-to-flight of longitudinal (axisymmetric) wave modes due to lamination in steel pipes" Proceedings of SPIE vol. 8695, Health Monitoring of Structural and Biological Systems 2013, 869515 (Apr. 17, 2013); 10 pgs. cited by applicant

Amjad, Umar et al., "Effects of transducers on guided wave based structural health monitoring" Proceedings of SPIE, vol. 10600, Health Monitoring of Structural and Biological Systems XII, 106000F (Apr. 23, 2018), 10 pgs. cited by applicant

Amjad, U. et al., "Generalized representations and universal aspects of Lamb wave dispersion relations" Proceedings of SPIE, vol. 7650, Health Monitoring of Structural and Biological Systems 2010, 76502F (Apr. 8, 2010), 9 pgs. cited by applicant

Amjad, Umar et al., "Detection and quantification of pipe damage from change in time of flight and phase" *Ultrasonics* vol. 62 (2015) pp. 223-236, Jun. 11, 2015, 14 pgs. cited by applicant

Amjad, Umar et al., "Detection and quantification of diameter reduction due to corrosion in reinforcing steel bars" *Structural Health Monitoring* 2015, vol. 14(5) 532-543, 12 pgs. cited by applicant

Amjad, Umar et al., "Detection and quantification of delamination in laminated plates from the phase of appropriate guided wave modes" *Optical Engineering* 55(1), Jan. 2016, 11 pgs. cited by applicant

API: American Petroleum Institute Preliminary Program, Oct. 16-17, 2019, 5 pages. cited by applicant

Gurkov, Andrey "Gigantic Druzhba oil pipeline paralyzed for weeks" May 7, 2019, 3 pages, <https://www.dw.com/en/gigantic-druzhba-oil-pipeline-paralyzed-for-weeks/a-48638989>. cited by applicant

Hassanzadeh et al., "Investigation of factors affecting on viscosity reduction of sludge from Iranian crude oil storage tanks", *Petroleum Science*, vol. 15, Jul. 2018, pp. 634-643. cited by applicant

Kak et al., "Principles of Computerized Tomographic Imaging", IEEE, 1988, Chapter 2, 48 pgs. cited by applicant

Luck, Marissa "Deer Park fire a 'blemish' for the petrochemical industry's image" *Houston Chronicle*, Mar. 26, 2019, 3 pages <https://www.houstonchronicle.com/business/energy/article/Deer-Park-fire-a-blemish-for-the-image-of-13717661.php>. cited by applicant

Pandey, "Ultrasonic attenuation in condensed matter", Dissertation for V.B.S. Purvanchal University, 2009, Chapter 1, 36 pgs. cited by applicant

Pepperl + Fuchs, "The Pulse of Automation", Technology Guide Ultrasonics by Pepperl + Fuchs, Germany 2017. Retrieved from <https://www.pepperl-fuchs.com/global/en/36955.htm>, 7 pgs. cited by applicant

Pluta et al., "Stress Dependent Dispersion Relations of Acoustic Waves Travelling on a Chain of

Point Masses Connected by Anharmonic Linear and Torsional Springs” *International Congress on Ultrasonics* AIP Conf. Proc. 1433, 471-474 (2012); 5 pgs. cited by applicant

Shelke, et al., “Mode-Selective Excitation and Detection of Ultrasonic Guided Waves for Delamination Detection in Laminated Aluminum Plates” *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, vol. 58, No. 3, Mar. 2011, 11 pgs. cited by applicant

“TOPS Terminal Operating Practices Symposium” Program Agenda, Apr. 11, 2018, 1 page. cited by applicant

Zadler, et al., “Resonant Ultrasound Spectroscopy: theory and application”, *Geophysical Journal International*, vol. 156, Issue 1, Jan. 2004, pp. 154-169. cited by applicant

Vermeersch, “Influence of substrate thickness on thermal impedance of microelectronic structures”, *Microelectronics Reliability*, 47, 2007, pp. 437-443. cited by applicant

Alpha Chemistry website, “Carbon nanotube sheet”, https://graphene.alfa-chemistry.com/product/carbon-nanotube-sheet-cas-308068-56-6-330280.html?nid=__, accessed Oct. 17, 2023, 6 pgs. cited by applicant

Attiya, “Lower Frequency Limit of Carbon Nanotube Antenna”, *Progress in Electromagnetics Research*, PIER 94, 2009, pp. 419-433, 16 pgs. cited by applicant

Ayub et al., “Acoustic absorption behaviour of carbon nanotube arrays”, *Inter-Noise 2014*, Melbourne Australia, Nov. 16-19, 2014, 10 pgs. cited by applicant

Calculla.com “Acoustic impedance of various materials table”, https://calculla.com/acoustic_impedance, accessed Oct. 18, 2023, 4 pgs. cited by applicant

Chico et al., “Pure Carbon Nanoscale Devices: Nanotube Heterojunctions”, *Physical Review Letters*, vol. 76, No. 6, Feb. 5, 1996, 4 pgs. cited by applicant

Ciplys, et al., “Attenuation of Surface Acoustic Waves by Carbon Nanotubes”, *Materials Research Society*, vol. 750, 2003, 6 pgs. cited by applicant

Cong et al., “Probing the acoustic phonon dispersion and sound velocity of graphene by Raman Spectroscopy”, *Carbon* 149, 2019, pp. 19-24, 6 pgs. cited by applicant

Delogu et al., “Functionalized multiwalled carbon nanotubes as ultrasound contrast agents”, *PNAS*, vol. 109, No. 41, Oct. 9, 2012, 6 pgs. cited by applicant

Dumitrica, et al., “Strain-rate and temperature dependent plastic yield in carbon nanotubes from ab initio calculations”, *Applied Physics Letters*, vol. 84, No. 15, Apr. 12, 2004, 3 pgs. cited by applicant

Eatemadi et al., “Carbon nanotubes: properties, synthesis, purification, and medical applications”, *Nanoscale Research Letters*, 2014, 13 pgs. cited by applicant

Engineering Toolbox website, “ASTM D3035—Polyethylene PE pipes—dimensions”, https://www.engineeringtoolbox.com/ASTM-D3035-PE-pipes-d_2138.html , accessed Oct. 17, 2023, 9 pgs. cited by applicant

Garcia-Gancedo et al., “AIN-based BAW resonators with CNT electrodes for gravimetric biosensing”, *Sensors and Actuators B: Chemical*, 2011, 8 pgs. cited by applicant

Huang et al., “Kink Formation and Motion in Carbon Nanotubes at Hight Temperatures”, *Physical Review Letters*, PRL 97, Aug. 18, 2006, 4 pgs. cited by applicant

Huang et al, “Superplastic carbon nanotubes”, *Nature*, vol. 439, Jan. 19, 2006, 1 pg. cited by applicant

Iborra et al., “Acoustic Properties of Carbon Nanotube Electrodes in BAW Resonators”, *IEEE*, 2013, pp. 984-987, 4 pgs. cited by applicant

Iijima et al., “Single-shell carbon nanotubes of 1-nm diameter”, *Nature*, vol. 363, Jun. 17, 1993, 3 pgs. cited by applicant

Iijima, “Helical microtubules of graphitic carbon”, *Nature*, vol. 354, Nov. 7, 1991, 3 pgs. cited by applicant

Kim et al., “Direct Observation of Localized Defect States in Semiconductor Nanotube Junctions”, *Physical Review Letters*, vol. 90, No. 21, May 30, 2003, 4 pgs. cited by applicant

Kozlov et al., “Sound of carbon nanotube assemblies”, J. Appl Phys. 105, 2009, Abstract only, 1 pg. cited by applicant

Magnin, et al., “Collapse phase diagram of carbon nanotubes with arbitrary number of walls. Collapse modes and macroscopic analog”, Carbon 179, 2021, pp. 552-562, 11 pgs. cited by applicant

Minary-Jolandan et al., “Reversible radial deformation up to the complete flattening of carbon nanotubes in nanoindentation”, Journal of Applied Physics, 103, 2008, 5 pgs. cited by applicant

Mori et al., “Energetics of plastic bending of carbon nanotubes”, Physical Review B 74, 2006, 5 pgs. cited by applicant

Nakayama et al., “Current-Induced Plastic Deformation of Double-Walled Carbon Nanotubes”, Japanese Journal of Applied Physics, vol. 44, No. 23, 2005, pp. L720-L722, 4 pgs. cited by applicant

NanoWerk website,
https://www.nanowerk.com/nanotechnology/introduction/introduction_to_nanotechnology_22.php
 accessed Oct. 17, 2023, 14 pgs. cited by applicant

Nardelli et al., “Mechanism of strain release in carbon nanotubes”, Physical Review B, vol. 57, No. 8, Feb. 15, 1998, 4 pgs. cited by applicant

NDTNet website, <https://www.ndt.net/links/proper.htm>. accessed Oct. 17, 2023, 2 pgs. cited by applicant

Ouyang et al., “Atomically Resolved Single-Walled Carbon Nanotube Intramolecular Junctions”, Science, vol. 291, No. 5501, Jan. 5, 2021, pp. 97-100, 5 pgs. cited by applicant

Palaci et al., “Radial elasticity of multi-walled carbon nanotubes”, Physical Review Letters, vol. 94, No. 17, 2005, 5 pgs. cited by applicant

PVC Pipe Dimensions—Schedule 40, <https://www.pvcfittingsonline.com/resource-center/pvc-pipe-dimensions-18-through-24/>, accessed Oct. 17, 2023, 2 pgs. cited by applicant

“Velocity Speed of Sound in Various Materials—RF Cafe”,
<https://www.rfcafe.com/references/general/velocity-sound-media.htm>, accessed Oct. 17, 2023, 3 pgs. cited by applicant

Rudoff et al., “Radial deformation of carbon nanotubes by van der Waals forces”, Nature, vol. 364, Aug. 5, 1993, 3 pgs. cited by applicant

Sato, “Elastic and Plastic Deformation of Carbon Nanotubes”, Procedia Engineering, 14, 2011, pp. 2366-2372, 7 pgs. cited by applicant

Shima, et al., “Chapter 6, Topological Defects”, Elastic and Plastic Deformation of Carbon Nanotubes, CRC Press, pp. 81-98, 2013, 18 pgs. cited by applicant

Signal Processing website, <https://www.signal-processing.com/table.php>, accessed Oct. 17, 2023, 3 ps. cited by applicant

TEPPFA, “Fast Guide to Materials”, <https://www.teppfa.eu/benefits-of-plastic-pipes-and-fittings/fast-guide-to-materials/>, accessed Oct. 17, 2023, 2 pgs. cited by applicant

Wikipedia “Carbon nanotube”, https://en.wikipedia.org/wiki/Carbon_nanotube, accessed Oct. 17, 2023, 35 pgs. cited by applicant

Wikipedia “Mechanical properties of carbon nanotubes”,
https://en.wikipedia.org/wiki/Mechanical_properties_of_carbon_nanotubes, accessed Oct. 17, 2023, 6 pgs. cited by applicant

Wikipedia, “Speed of Sound”, https://en.wikipedia.org/wiki/Speed_of_sound, accessed Oct. 17, 2023, 19 pgs. cited by applicant

Yang et al., “Radial elasticity of single-walled carbon nanotube measured by atomic force microscopy”, Applied Physics Letters, 98, 2011, 3 pgs. cited by applicant

Yoon et al., “Sound wave propagation in multiwall carbon nanotubes”, J. Appl. Phys. 93, 2003, Abstract only, 1 pg. cited by applicant

Yu, et al., “Strength and Breaking Mechanism of Multiwalled Carbon Nanotubes Under Tensile

Load”, Science, vol. 287, Jan. 28, 2000, 4 pgs. cited by applicant
Yu, et al., “Investigation of the Radial Deformability of Individual Carbon Nanotubes under Controlled Indentation Force”, Physical Review Letters, vol. 85, No. 7, Aug. 14, 2000, 4 pgs. cited by applicant
Zhang et al., “Plastic Deformations of Carbon Nanotubes”, Physical Review Letters, vol. 81, No. 24, Dec. 14, 1998, 4 pgs. cited by applicant
Zhou et al., “Formation energy of Stone-Wales defects in carbon nanotubes”, Applied Physics Letters, vol. 83, No. 6, Aug. 11, 2003, 3 pgs. cited by applicant

Primary Examiner: Saint Surin; Jacques M

Attorney, Agent or Firm: HAYES SOLOWAY P.C.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 18/223,987 filed Jul. 19 2023, which in turn claims benefit of U.S. Provisional Application Ser. No. 63/368,856 entitled, “Acoustic Signal Material Identification with Nanotube Couplant” filed Jul. 19, 2022, the entire disclosure of which is incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure is generally related to acoustic signal material detection, and more particularly is related to acoustic signal material identification with nanotube couplant.

BACKGROUND OF THE DISCLOSURE

(2) Pipes and pipelines are commonly used in a variety of industries to transport fluids. For instance, water pipes transport potable and sewer water in urban areas, pipes are used to transport chemicals within factories, and pipelines are used within the oil and gas industry for transporting petroleum products within refineries or between various locations. Similarly, various containers and vessels are often used to store fluids, such as oil storage tanks, chemical storage vessels, and the like. To determine the type of fluid within the vessel, container, pipes, or pipelines, acoustic-based sensors can be used to determine characteristics of the fluid which can be used to then identify the material type or material identity of the fluid.

(3) Acoustic signals are commonly used for determining material identity of fluids, or otherwise assessing fluids and other materials within containers and pipelines, including those used to store oil and gas within the petroleum industry. There are many reasons to use acoustic waves for measurements of fluids or materials in a container or other type of enclosure. For instance, some containers are not easily accessible, such as underground storage tanks and large, multi-story fuel storage containers. Acoustic waves for measurements are also especially useful for metal enclosures and other non-transparent enclosures that encapsulate potentially hazardous materials, such as oil, gas, fuel, and various chemicals at different temperatures. These may be prevalent in processing plants, chemical plants, food industry plants, nuclear power stations, power grid transformers, and refineries.

(4) However, despite the ability to use acoustic signals for determining material identity and otherwise assessing fluids and materials, such an approach is often complicated by faulty or subpart acoustic signal transmissions. In many cases, the acoustic signal transmission may be weakened or altered based on the surrounding environment or the materials within the acoustic signal transmission path. When acoustic signal transmissions are faulty or inaccurate, they lead to inaccurate readings which provide inaccurate results.

(5) Thus, a heretofore unaddressed need exists in the industry to address the aforementioned deficiencies and inadequacies.

SUMMARY OF THE DISCLOSURE

(6) Embodiments of the present disclosure provide a system for improved, non-intrusive material identification. Briefly described, in architecture, one embodiment of the system, among others, can be implemented as follows. A vessel is holding or transporting at least one quantity of a fluid. At least one acoustic transducer is positioned on an exterior surface of a sidewall of the vessel. At least one coupling layer is positioned between the at least one acoustic transducer and the exterior surface of the sidewall of the vessel, wherein the at least one coupling layer is formed, at least in part, from nanotubes, wherein the nanotubes improve a reflection of an acoustic signal emitted from the at least one acoustic transducer. A computerized device is in communication with the at least one acoustic signal, the computerized device having a processor and a memory, wherein the computerized device determines a material identification of the quantity of fluid based on, at least in part, the reflection of the ultrasonic signal.

(7) The present disclosure can also be viewed as providing methods for improved, non-intrusive material identification. In this regard, one embodiment of such a method, among others, can be broadly summarized by the following steps: holding or transporting at least one quantity of a fluid in a vessel; positioning at least one acoustic transducer on an exterior surface of a sidewall of the vessel; positioning at least one coupling layer between the at least one acoustic transducer and the exterior surface of the sidewall of the vessel, wherein the at least one coupling layer is formed, at least in part, from nanotubes, wherein the nanotubes improve a reflection of an acoustic signal emitted from the at least one acoustic transducer; and using a computerized device in communication with the at least one acoustic signal, the computerized device having a processor and a memory, determining a material identification of the quantity of fluid based on, at least in part, the reflection of the ultrasonic signal.

(8) Other systems, methods, features, and advantages of the present disclosure will be or become apparent to one with skill in the art upon examination of the following drawings and detailed description. It is intended that all such additional systems, methods, features, and advantages be included within this description, be within the scope of the present disclosure, and be protected by the accompanying claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the present disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

(2) FIG. 1 is a diagrammatical illustration of a system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(3) FIG. 2 is a diagrammatical illustration of a system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(4) FIG. 3 is a diagrammatical view illustration of the system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(5) FIG. 4A is a diagrammatical view illustration of a single-walled carbon nanotube used with the system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(6) FIG. 4B is a diagrammatical view illustration of a multi-walled carbon nanotube used with the system for non-intrusive material identification, in accordance with exemplary embodiments of the

present disclosure.

(7) FIG. 5 is a diagrammatical view illustration of the system for non-intrusive material identification without a couplant, in accordance with exemplary embodiments of the present disclosure.

(8) FIG. 6 is a diagrammatical view illustration of a system for non-intrusive material identification without a couplant, in accordance with exemplary embodiments of the present disclosure.

(9) FIG. 7 is a diagrammatical view illustration of a system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(10) FIG. 8A is a diagrammatical view illustration of a system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(11) FIG. 8B is a diagrammatical view illustration of a system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(12) FIG. 9A is a diagrammatical view illustration of a system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(13) FIG. 9B is a diagrammatical view illustration of a system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(14) FIG. 10 illustrates a diagrammatical view illustration of a system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure.

(15) FIG. 11 illustrates an example of the subject disclosure, in accordance with exemplary embodiments of the present disclosure.

DETAILED DESCRIPTION

(16) In the current era of nanotechnology, a variety of nanotube constituents are available which can be formed into nanoscopic fibers, for instance: carbon nanotubes or nanotubes created from other material or their composites. Advances in nanotechnology have provided acoustic researchers with a number of new materials with nanofibers and nano-pores that can potentially be used in transducer design and as coupling layers. Multiple composite materials were created to explore the capabilities on nanotubes which are waiting for utilization in various industries.

(17) The extraordinary thermal and elastic properties of graphene have attracted great attention in its fundamental research and practical applications. Most of the physical properties of carbon nanotubes (CNTs) derive from graphene. In graphene, carbon atoms are densely organized in a regular bonded atomic-scale honeycomb (hexagonal) pattern, and this pattern is a basic structure for other carbon bonded materials (allotropes) such as fullerenes and carbon nanotubes. Carbon nanotube is a cylinder fabricated of rolled up grapheme sheet. Nanotubes can be single-walled carbon (and/or other materials) nanotubes (SWCNTs) and multiwalled carbon (and/or other materials) nanotubes (MWCNTs). Multi-walled nanotubes are multiple concentric nanotubes precisely nested within one another, which can be aligned in one direction and implemented as an array. The MWCNTs are just connected with Van der Waals forces, and no binder material is needed.

(18) The CNT's have high anisotropic transport properties, including electrical conductivity and thermal conductivity. CNT hexagon two-dimensional structure shows high level of crystal quality. MWCNT show in addition striking telescoping property whereby an inner nanotube core may slide, almost without friction, within its outer nanotube shell, thus creating an atomically perfect linear or rotational motion that shows nearly no friction.

(19) Chemical reactivity is a huge impetus for material identification applications where chemical substances undergo chemical reactions and this is usually coupled with an overall release of significant energy leading to extreme temperatures in the reactor itself or the walls of the container encapsulating the reactor. This may allow for material identification applications in harsh environments. The CNT specific strength is several orders of magnitude higher than steel as versus the stainless steel 63.1 kN.Math.m.Math.kg.sup.-1 or high-carbon steel 154

kN.Math.m.Math.kg.sup.-1. Temperatures that CNT's can withstand are until 2500 F and 1000 F,

although in vacuum their resistance is close to the theoretical 4000 F. The CNT's can transmit a wide range of frequencies without significant absorption.

(20) The unique properties of the carbon nanotubes make them ideal material for identification of material properties in multi-layered applications. Both single walled CNTs as well as CNTs with multi-walled structures may be used. MWCNTs are components of acoustic sensor design in the non-destructive material properties analysis and more scenically but not limited to ultrasonic testing. This approach presents implementation of the emerging molecular nanotechnology where precise positioning of atoms is applied to create useful machines and allows for exploration of quantum effects combined with electrical and mechanical dynamic of the nano material in measurements of the target material properties and environmental conditions like density and temperature as one example.

(21) Although a number of applications of CNTs in noise management have been suggested their acoustic properties have not been widely studied. Nano technology enabled ultrasonic transducer and couplant design for material identification applications with wide frequency range and low energy loss that are capable of withstanding substantial heat and virtually no chemical reactivity for applications in harsh environments may be used, but many other applications are envisioned.

(22) CNT/MWCNT arrays will guide the development of an effective acoustic transducers and coupling layers which will make use of various arrangements and compositions of (carbon, non-carbon, or composite) nanotube arrays.

(23) The strength and flexibility of carbon nanotubes make them of potential use in controlling other nanoscale structures, which suggests they will have a significant role in nanotechnology engineering. Carbon can bond in different ways to construct structures with completely different properties. This may enable transducers to bond to materials without an added coupling.

(24) Further efforts to quantify the acoustic characteristics and absorption performance of nanoscopic fibers and acoustic absorption coefficient of vertically aligned carbon nanotube (CNT) arrays will improve the design of coupling layers and transducers.

(25) Typical piezoelectric ultrasonic transducer under free conditions loose significant power when converting electrical energy to mechanical due to the contributions of the mechanical and electric power losses. The losses in a Langevin ultrasonic transducer can be divided into three parts: dielectric loss of piezoelectric, structural damping loss of the transducer's mechanical components, and losses due to the friction of contact surfaces. Estimate dielectric losses of the piezoelectric ceramic can be measured directly. Dielectric power loss is obtained by calculating the electrical energy stored in a piezoelectric and using the loss factors. The structural damping of the mechanical components of the transducer is calculated using the mechanical quality factors of each component. The factors have been measured using the standard modal tests. Structural power losses are obtained by calculating the elastic energy stored in each component and using the mechanical quality factors. Based on the calculations for the transducer with the resonance frequency of 20 kHz, under free conditions, the input power is 532 W, the total structural power loss at the resonance frequency is 221.82 W, the dielectric power loss is 15 W, and the power loss due to friction is 184.18 W. In this example nearly 80% of the energy is lost and in practice the losses can be higher due to using different frequencies in addition to impedance acoustic mismatch. Bulk of the losses in the above example are in the resonating element, therefore any improvement of the material used to generate (or convert from electrical) the mechanical waves will contribute to significant improvement of signal output without using additional input energy.

(26) These losses can be mitigated by the use of multiwalled nanotubes of any type of material where the internal tubes slide nearly without any friction. Kinetic properties of single- and multi-walled nanotubes (like CNT's or MWCNT) or flat two-dimensional sheets (Graphene in the case of pure carbon) of nanomaterials will serve dual purposes of filters to reduce the noise or remove unwanted signals from the receiving transducer and eliminate unwanted signals when reflecting from the outside wall surface. Thus, improving the noise to signal ratio or selecting specific signals

to measure. This way it is possible to select the information desired by selecting an appropriate nano technology enabled material and designing the appropriate internal metamaterial architecture with desired properties that will serve also as couplant in addition to transducer. They will be either implemented as a simple solution or in part of multilayered coupling or multipart transducer solution.

(27) One part of any couplant or transducer design which may be important is the impedance of the material that is being investigated, such as, for example, the impedance of the outside layer of a container. When a multi-layered material is investigated, the maximum signal energy in a specific layer may depend on the losses in the surrounding layers. Therefore, it may be prudent to consider the system of material in its entirety to resolve the issue of signal strength.

(28) Couplant layers can be assembled from nanomaterials in many forms like multi-layered graphene sheets of other two-dimensional single layer atomic sheets of mixed types of atoms and molecules that produce the correct impedance, adhesiveness and material mechanical properties needed for the couplant is a specific application. Additionally, nano tubes, nano spheres from different materials or mixtures can be used to create need impedance and mechanical response.

(29) Multiple layers of materials may often perform better than a signal layer where the ideal material interface for one couplant layer or multiple layers can be calculated. In the case of multiple layers, it may be possible to perfect for all the layers at the same time. This optimization may start from the top layer and continue down to the last layer, as the perfect match for both layers is calculated in each step. It may be possible to change the number of layers, and the size of the step in terms of impedance may change during the optimization algorithm in the method described herein.

(30) The layer(s) in the couplant can be mixture from nano components an any other materials including gels including but not limited to some of the existing couplants on the market.

(31) The multiple layers may or may not be only from nano-materials or materials enhanced with nano materials in any form, i.e., randomly distributed through the host material of oriented in some fashion.

(32) The subject disclosure is directed to the nanotube enabled systems and methods for improved transducer and couplant performance in ultrasonic applications for material identity detection and other material characteristic assessment. FIG. 1 is a diagrammatical illustration of a system for non-intrusive material identification **10**, in accordance with exemplary embodiments of the present disclosure. The system for non-intrusive material identification **10**, which may be referred to as 'system **10**' includes a vessel **20** holding or transporting at least one quantity of a fluid **22**. As shown, the vessel **20** is depicted as a pipeline having a fluid **22** which may move through the vessel **20** in one or more directions, but the vessel **20** may also include any other type of fluid containment or transportation device or system. For instance, the vessel **20** may be a storage tank, a container, a network of pipes, a holding facility, or another similar structure. As shown in FIG. 1, the vessel **20** includes a sidewall **24** having an interior surface **26** which forms the interior passageway or compartment for fluid **22** storage or transportation. An exterior of the sidewall **24** is defined by an exterior surface **28** which is positioned opposite the interior surface **26** across the sidewall **24**. Commonly, the exterior surface **28** may be exposed to the surrounding environment in which the vessel **20** is located. The exterior surface **28** may be encapsulated or covered, or it may be directly exposed to a surrounding environment.

(33) The fluid **22** within the vessel **20** may include any type of fluid, such as, for instance, oil, gas, fuel, a chemical, water, a composition of materials, or any other type of material or mix of materials. In this disclosure, oil and gas are used as exemplary fluids **22** to provide clarity in disclosure.

(34) At least one acoustic transducer **30** is positioned on an exterior surface **28** of the sidewall **24** of the vessel **20**. The acoustic transducer **30** may be an acoustic or ultrasonic sensing device which is capable of emitting one or more acoustic or ultrasonic signals **32**, e.g., an ultrasonic wave with

particular characteristics, and may be capable of receiving one or more acoustic or ultrasonic signals **32**. As shown in FIG. 1, the acoustic signals **32** may include a signal emitted from the acoustic transducer **30** into the vessel **20** and a reflection of that emitted signal **32** back to the acoustic transducer **30**. Any number of acoustic transducers **30** may be included on the vessel **20**, and the acoustic transducers **30** may be positioned in various locations or patterns. For instance, it may be possible for acoustic transducers **30** to be positioned in a linear configuration along a vessel **20**, such as horizontally along a pipeline or vertically along a storage vessel. In another example, acoustic transducers **30** may be positioned axially about a vessel **20**, spiraled about a vessel **20**, positioned at substantially equidistant locations about a perimeter or circumference of a vessel **20**, or in another configuration, all of which are considered within the scope of the present disclosure.

(35) Acoustic signals **32** are capable of being emitted into a sidewall **24** of a vessel **20** by the acoustic transducer **30**, whereby the signals **32**, or a portion thereof, move through the sidewall **24** proximate to the mounting location of the acoustic transducer **30**, and into the fluid **22** within the vessel **20**. All or a portion of the signals **32** may contact the opposing sidewall **24**, e.g., distal or opposite from where the acoustic transducer **30** is mounted, where all or a portion of those signals **32** are reflected back to the acoustic transducer. The acoustic transducer **30**, or another acoustic transducer **30**, may sense the reflected signal.

(36) At least one computerized device **50** may be in communication with the acoustic transducer **30**, or any portion of a plurality of acoustic transducers, such that the computerized device **50** can receive the acoustic signal **32** or data derived from the acoustic signals **32**. The computerized device **50** may include any computer having a memory **52** and a processor **54** which is capable of executing instructions for performing functions. The computerized device **50** may be in communication with the acoustic transducer **30** through any type of communication network, such as a wired or wireless network, direct or indirect, as well as be in communication with other devices, such as other computers, other networks, the Internet, or other computing architecture. The computerized device **50** may determine the material identity of the fluid **22** within the vessel **20**, or otherwise assess or determine a characteristic or property of the fluid **22** or the sidewall **24** of the vessel, based at least in part on the acoustic signal **32**.

(37) To improve signal transmission into the sidewall **24** of the vessel **20**, at least one coupling layer **40** may be positioned between the at least one acoustic transducer **30** and the exterior surface **28** of the sidewall **24** of the vessel **20**. The couplant **40** may be formed, at least in part, from nanotubes, which may include nano particles or similar nano-materials, whereby the nanotube couplant improves reflection of the signal **32** emitted from the at least one acoustic transducer **30**. In one example, the nanotube couplant may be formed from a couplant material with the addition of a nanotube materials, and optionally, other materials. The nanotube couplant **40** may help ensure that the system with the acoustic transducer **30** can be used for material identification applications with wide frequency range and less energy loss. Additionally, the nanotube couplant **40** may allow the acoustic transducer **30** to be capable of withstanding substantial heat and low chemical reactivity for applications in harsh environments.

(38) In use, most materials which receive acoustic signals, despite being tactically smooth, are often very uneven on their surfaces, such that there are numerous inequalities, pits, cavities, or textures on the surface. On a micro-level, these inequalities trap air and other materials, which reduce the surface through which signals can be transmitted. The use of the nanotube couplant **40** acts to fill in these inequalities and gaps, such that there is a more homogeneous surface and material through which signals can be sent, which in turn, allows for signals to pass with greater energy. Building single or multiple layer nano particles for a nano layer may allow for the nano particles to coat the uneven parts of each surface so it is possible to use middle layer or layers for effective energy transfer among rigid objects.

(39) The nanotube couplant **40** can be used in various different configurations. For example, as shown in FIG. 1, a nanotube couplant **40** may be positioned between the acoustic transducer **30** and

the sidewall **24** of the vessel **20** in a location proximate to just that acoustic transducer **30**. In this configuration, the signals **32** are emitted through the nanotube couplant **40**, which helps ensure there is less energy loss in the signal transmission through the sidewall **24**, as compared to non-nanotube couplants or no couplant at all.

(40) In FIG. 2, which is a diagrammatical illustration of a system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure, a second couplant **42** is used on an opposing sidewall **24** from the emitting transducer **30**. This second couplant **42** may be used as a reflection layer to the signals **32** transmitted from the acoustic transducer **30**, and may act to improve signal reflection, i.e., transmission of that signal through the opposing portion of the sidewall **24** of the vessel **20** and reflected back to the acoustic transducer **30**. It is noted that in FIG. 2, the signal **32** may reflect off of the second nanotube couplant **42**, whereas in the example of FIG. 1, the signal **32** reflects off the inner surface **26** of the sidewall **24** of the vessel **20**.

(41) In another example, FIG. 3 is a diagrammatical view illustration of the system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure, where a second nanotube couplant **42** and a second acoustic transducer **34** are both used. This configuration allows for improved pitch-catch signal transmission, where signal **32** is emitted from the first transducer **30** and is received by the second transducer **34** to achieve a pulse-echo configuration. Here, each of the first and second nanotube couplants **40**, **42** help prevent energy loss of the signal **32** when transmitted through the sidewall **24** of the vessel **20**. The signal **32** may be able to reflect off of the opposing nanotube couplant **42**, as shown, or they may be configured to reflect off of the inner surface **26** of the sidewall **24** of the vessel **20**.

(42) Other configurations for using one or more nanotube couplants **40** with one or more acoustic transducers **30** are also contemplated, all of which are considered within the scope of the present disclosure.

(43) Recent advances in nanotechnology have yielded a number of new materials with nanofibers and nano-pores that can be implemented in acoustic applications. Despite isolated measurements of the acoustic absorption properties of nano materials having not been studied widely, it has been discovered that applications of the acoustic properties presented from carbon nanotubes (CNT), or multi-walled carbon nanotubes (MWCNT) arrays may provide benefits with designing effective acoustic transducers and coupling layers which may make use of various arrangements and compositions of nanotube arrays, including carbon, non-carbon or composite. Thus, the use of nanotube couplants **40** in any configuration can provide substantial improvements in preventing signal energy loss, in allowing a wide frequency range of signals to be used, and in withstanding substantial heat and low chemical reactivity for applications in harsh environments, among other benefits.

(44) Using carbon nanotubes as an exemplary material, it is noted that the strength and flexibility of carbon nanotubes make them of potential use in controlling other nanoscale structures, which suggests they will have a significant role in nanotechnology engineering. Most of the physical properties of carbon nanotubes derive from graphene. In graphene, carbon atoms are densely organized in a regular sp²-bonded atomic-scale honeycomb (hexagonal) pattern, and this pattern is a basic structure for other sp² carbon bonded materials (allotropes) such as fullerenes and carbon nanotubes. Carbon nanotubes are theoretically distinct as a cylinder fabricated of rolled up grapheme sheet. It can be divided into a single well or multiple wells.

(45) Nanotubes with a single well are described as single-wall carbon nanotubes (SWCNTs), while the ones with more than one well are MWCNTs. FIG. 4A is a diagrammatical view illustration of a single-walled carbon nanotube used with the system for non-intrusive material identification, in accordance with exemplary embodiments of the present disclosure, whereas FIG. 4B is a diagrammatical view illustration of a multi-walled carbon nanotube used with the system for non-intrusive material identification, in accordance with exemplary embodiments of the present

disclosure. Carbon can bond in different ways to construct structures with completely different properties. This will enable transducers to bond to materials without an additional coupling.

(46) As shown in FIG. 4B, MWCNTs are multiple concentric nanotubes precisely nested within one another. One potential example of these are MWCNT that can be aligned in a direction and implemented in an array. The MWCNTs are just connected with van der Waals forces, and no binder material is needed. These show a striking telescoping property whereby an inner nanotube core may slide, almost without friction, within its outer nanotube shell, thus creating an atomically perfect linear or rotational motion that shows nearly no friction.

(47) Both SWCNTs as well as MWCNTs may be used within the nanotube couplant **40** of FIGS. 1-3. MWCNTs may be used with substantial benefits as a couplant or as components of acoustic sensor design for providing non-intrusive or non-destructive material properties analysis, including material identification or assessment of fluids within vessels **20**, or with other forms of material analysis using ultrasonic testing. The system **10**, as described herein may utilize emerging molecular nanotechnology where precise positioning of atoms is applied to create improvements in acoustic sensing, including material identification applications with wide frequency range and less energy loss and capable of withstanding substantial heat and low chemical reactivity for applications in harsh environments.

(48) In chemistry, reactivity is a huge impetus for material identification applications where chemical substances undergo chemical reactions. This is usually coupled with an overall release of significant energy leading to extreme temperatures in the reactor itself or the walls of the container encapsulating the reactor. For instance, the extraordinary thermal and elastic properties of graphene, mainly originating from its unique acoustic phonon branches near Γ point in Brillouin zone, have attracted great attention in its fundamental research and practical applications. The CNT's have high anisotropic transport properties, including electrical conductivity and thermal conductivity. Crystal quality of CNT has been confirmed by Raman scattering measurement. Intensity ratio of G-peak (1580 cm^{-1}) and D-peak (1350 cm^{-1}) represents quality of graphitization of carbon materials. G/D of CNT is ~ 3 , showing high crystal quality. The high crystallinity may also be supported by transmission electron microscopy.

(49) With the use of nanotube couplants **40**, acoustic signal transmission through walls of vessels **20** or other containment units can be improved, which can increase the accuracy of material type identification in the fluid within the vessel, where the signals **32** are used, at least in part, to make a determination of a material identification of the quantity of fluid using known techniques. It is noted that the nanotube material forming the couplant, in whole or part, may include various characteristics or designs. For instance, it may be possible to apply single or multiwalled nanotubes of any material as a coupling layer between the acoustic transducer **30** and the material which is desired to be analyzed or investigated, such as by applying a layer of carbon nanotubes on the outside of the vessel **20** or pipe to improve reflection of the ultrasound wave in multiple conditions. A standard transducer may be used with one or more nanotube coupling layers to improve existing transducers, thereby allowing existing transducer technology to operate better than it conventionally does. It may also be possible to apply one or more coupling layers, which may allow for future applications, such as a gigahertz mechanical oscillator where there are no losses in the transducer. It is also possible to use vertically aligned nanotube arrays to improve existing mechanical properties of the transducers, including with the speed of sound, impedance, and attenuation.

(50) Tables 1 and 2, below, illustrate supporting data to the improvement of impedance by using nanotubes in couplants relative to various other materials:

(51) TABLE-US-00001 TABLE 1 Impedance CNT density $d\ 1.3\text{ g/cm}^3$ 13 kg/m^3 speed $s\ 12000\text{ m/s}$ 12000 m/s Impedance $d*s\ 156,000.00\text{ kg/s} \cdot \text{m}^2$ CNT reference Impedance density $d\ 1.3\text{ g/cm}^3$ 6500.00 kg/m^3 speed $s\ 12000\text{ m/s}$ 12000.00 m/s density $d\ 1.3$

ρ /cm³ 866.67 kg/m³ speed v 9000 m/s 9000 m/s
 Impedance $d \cdot v$ 78,000,000.00 kg/s .Math. ρ CNT density d 1.3
 ρ /cm³ 13 kg/m³ speed v 90000 m/s 90000 m/s
 Impedance $d \cdot v$ 1,170,000.00 kg/s .Math. ρ Steel m/s Longitudinal density
 7980 kg/m³ speed 5900 m/s Impedance 47,082,000.00 kg/s .Math.
 ρ Sheer density 7980 kg/m³ speed 3600 m/s
 Impedance 28,728,000.00 kg/s .Math. ρ PZT Ceramic density d 750
 ρ /cm³ 7500 kg/m³ speed v 3800 m/s 3800 m/s
 Impedance $d \cdot v$ 28,500,000.00 kg/s .Math. ρ Quartz density d 750
 ρ /cm³ 2600 kg/m³ speed v 5700 m/s 5700 m/s
 Impedance $d \cdot v$ 14,820,000.00 kg/s .Math. ρ Glycerin density d 1.26
 ρ /cm³ 1260 kg/m³ speed v 1904 m/s 1904 m/s
 Impedance $d \cdot v$ 2,399,040.00 kg/s .Math. ρ Echo Z density d 1.4
 ρ /cm³ 1400 kg/m³ speed v 2600 m/s 2600 m/s
 Impedance $d \cdot v$ 3,640,000.00 kg/s .Math. ρ Aluminum d g/cm³
 ρ kg/m³ kg/s .Math. ρ s m/s m/s $d \cdot v$ Glass d
 ρ /cm³ kg/m³ kg/s .Math. ρ s m/s
 $d \cdot v$ Polymer density d 180 g/cm³ 1800 kg/m³ speed
 v 2200 m/s 2200 m/s Impedance $d \cdot v$ 3,960,000.00 kg/s .Math. ρ Sonotech:
 Ultra gel II d g/cm³ 1090.909 kg/s .Math. ρ s 1650 m/s
 1650 m/s $d \cdot v$ 1,800,000.00 kg/s .Math. ρ Water density d 1 g/cm³
 ρ 1000 kg/m³ speed v 1480 m/s 1480 m/s Impedance $d \cdot v$
 1,480,000.00 kg/s .Math. ρ Air density d 0.001204 g/cm³
 ρ 1.2041 kg/m³ speed v 343.21 m/s 343.21 m/s Impedance $d \cdot v$
 413.26 kg/s .Math. ρ Methane density d 0.00108 g/cm³ 1.08
 kg/m³ speed v 415 m/s 415 m/s Impedance $d \cdot v$ 448.20 kg/s .Math.
 ρ PZT density d 0.00108 g/cm³ 1.08 kg/m³ speed
 v 415 m/s 415 m/s Impedance $d \cdot v$ 448.20 kg/s .Math. ρ
 (52) TABLE-US-00002 TABLE 2 Optimal Reflection Transmission Optimal Layer $((Z_2 - Z_1)/$
 $4 \cdot z_2 \cdot z_1 /$ Transmission $Z_1 Z_2$ Impedance Thickness $(Z_2 + Z_1)) \cdot (z_2 + z_1)$
 ρ [%] Steel/Sheer PZT Ceramic -0.23 57.23 28.73
 28.50 28.61 0.0015872764% 99.9984127236% 99.99841272% PZT Ultrigel II
 -26.70 30.30 Ceramic 28.50 1.80 77.6492500735% 22.3507499265%
 22.35074993% Ultrigel II Steel/Sheer 28.61 1.80 28.73 77.8057137771%
 22.1942862229% 4.96058941% Quartz PZT Ceramic 14.82 28.5
 9.9722991690% 90.0277008310% 90.02770083% PZT Ultrigel II Ceramic 28.50 1.80
 77.6492500735% 22.3507499265% 20.12186628% Ultrigel II Steel/Sheer 28.61 1.80
 28.73 77.8057137771% 22.1942862229% 4.46590459% Steps softer 1 PZT
 Ultrigel Optimal Reflection Transmission Transmission than CNT Ceramic II 28.50 1.80
 77.65% 22.35% 22.35% 2 Ultrigel II Steel/Long 36.63 1.80 47.08 85.81%
 14.19% 3.17% Stainless Air Reflection Transmission Transmission Sheer
 28.73 0.000448200000 99.9937595932% 0.0062404068% 0.00624041% Air Stainless
 Steel 0.000413259 28.728 99.9942460697% 0.0057539303% 0.00575393%
 Stainless CNT Long 47.08 78.00 6.1098866330% 93.8901133670%
 93.89011337% CNT Air 78.00 0.000413259161 99.9978807447% 0.0021192553%
 0.00198977% Steps 1 PZT CNT Optimal Reflection Transmission Transmission Ceramic 28.50
 78.00 21.60% 78.40% 78.40% 2 CNT Stainless 36.63 Long 78.00 47.082
 6.11% 93.89% 73.61%

(53) Tables 3 and 4 illustrate practical examples of the benefit of the system **10**. In particular, Table 3 illustrates traditional transducer using (PZT Ceramic) and couplant (Ultrigel II) to interface with

steel wall:

(54) TABLE-US-00003

TABLE 3	Material/	Material/	Signal after	Signal after	Signal in Steps
Impedance	Impedance	Optimal	Reflection	Transmission	the Container
1	PZT Ceramic	Ultragel II	Impedance 28.50	1.80	77.65%
2	Ultragel II	Steel/Long	36.63	1.80	47.08
					85.81%
					14.19%
					3.17%

This table shows that traditional transducers loose potentially ~90%+ of the initial signal in the transducer only, and there is evidence that 80% of these losses are centered in the conversion of electric to mechanical energy.

(55) Table 4 illustrates data in an example where a transducer using (PZT Ceramic) and a nano base couplant (CNT) to interface with the wall is used:

(56) TABLE-US-00004

TABLE 4	Material/	Material/	Signal after	Signal after	Signal in Steps
Impedance	Impedance	Optimal	Reflection	Transmission	the Container
1	PZT Ceramic	CNT	Impedance 28.50	78.00	21.60%
2	CNT	Stainless Long	36.63	78.00	47.082
					6.11%
					93.89%
					73.61%

Here, the nanotubes material is using “standard” maximum impedance, however it is possible to find appropriate design and material composition to achieve the target impedance for this case of 36.63 and therefore 100% signal transmission. For Tables 3-4, all impedance values are in MRayls units.

(57) FIG. 5 is a diagrammatical view illustration of a system for non-intrusive material identification without a couplant **110**, in accordance with exemplary embodiments of the present disclosure. In particular, the system **110** depicts the use of a pulse-echo scenario of nanotubes-based transducer **130** that would not require additional interface layer. The nano technology based transducer **130** may not need the couplant, as described relative to FIGS. 1-4, since the impedance can be adjusted for the signal **132** to be transmitted and/or received to achieve the desired results, which can be communicated to the computerized device **150** for material identification. The system **110** of FIG. 5 using pulse-echo may require only one nano enabled transducer **130**. The benefits of the nano technology in this transducer **130** can be used once for the full cross pipe roundtrip. This may allow for applications that can be served by just minimum improvements of the signal strength.

(58) FIG. 6 is a diagrammatical view illustration of a system for non-intrusive material identification without a couplant **110**, in accordance with exemplary embodiments of the present disclosure. In this situation, it may be possible use a setup that includes two transducers **130**, however only one is sending the signal **132** and both transducers **130** are recording the echoes. This may make it possible to use the pulse-echo and pitch-catch echoes techniques at the same time. This may provide the opportunity to sense how the echoes decay in time where the signal **132** is crossing the pipe **120** only once. In pitch-catch, two nano-based transducers may derive significant signal improvement benefit i.e. the signal will be with a higher signal to noise ratio and twice more frequent measurements since it may be possible to use both transducers for measurements.

(59) In both FIGS. 5-6, the nanotube couplant of FIGS. 1-3 may not be needed because the nanotube technology is incorporated into the sensing portion of the transducer(s) **130** itself, such that the nanotubes can be used as part of the sensor. When the nanotubes are used as part of the sensor, even the smallest signal can be transferred to the nanotube sensor which can register that signal. This is especially true for multi-walled nanotubes, since the nanotubes are nested in other tubes, the inner tubes will move or raise up when any signal is received. This creates a vibration medium that does not create friction, such that even the slightest signals can be detected.

(60) It is also noted that a nanotube couplant can be used with FIGS. 5-6 to further enhance the coupling to the vessel, where the nanotube couplants can be used to fill uneven surfaces within the surfaces of the vessel.

(61) FIG. 7 is a diagrammatical view illustration of a system for non-intrusive material identification **210**, in accordance with exemplary embodiments of the present disclosure. As shown

in FIG. 7, nano technology may be used for assisted signal 232 transmission from transducers 230 within the pipe or vessel wall 220 bouncing for enhanced material detection, even with a material 222 or fluid within the pipe. Multiple bounces within the wall 220 may facilitate multiple interaction with the material 222 however the interaction with the air is an unknown variable to an extent since the exact air composition is not known. As shown in FIG. 7, the outside air contact is effectively replaced with one or more nano technologies enabled layers 240 which has known impedance, which is positioned between the transducers 230 on the exterior sidewall of the vessel wall 220. An acoustic signal 232 emitted from one or both of the transducers 230 may reflect between the inner and outer surfaces of the sidewall 220 and the nano technologies enabled layers 240. In this arrangement, it may be possible to predict the material inside the container better since the impedance of the nano technologies enabled layers 240 is known, whereas the ambient air or atmosphere may not be known.

(62) FIG. 8A is a diagrammatical view illustration of a system for non-intrusive material identification 310, in accordance with exemplary embodiments of the present disclosure. As shown, FIG. 8A depicts a couplant layer design with oriented array of single- or multiwalled-nanotubes 320 either filled with air or other materials to additionally change the impedance. The nanotubes 320 may form a vertically aligned nanotube array with each nanotube positioned substantially axially parallel to the other nanotubes, or an array aligned in a different orientation. This nanotube oriented array couplant design may provide improved signal transmission and noise range filtering/selection as well as high-temperature applications, low signal losses and wide frequency range of signal transmission enabling for better detection of materials inside the pipe/container. The nano materials can vary widely that would lead to implementation of wide range of impedance for single or multiple layer matching for signal improvement.

(63) FIG. 8B is a diagrammatical view illustration of a system for non-intrusive material identification 310, in accordance with exemplary embodiments of the present disclosure. As shown, FIG. 8B illustrates graphene-like sheets with a nearly two-dimensional shape which can be oriented perpendicular or horizontal in relation to the surface of the wall of the container. The graphene-like sheets may be positioned substantially parallel to one another along a planar orientation of the sheets. These multiple nanotube graphene-like sheets couplant designs may provide improved signal transmission and noise range filtering/selection as well as high-temperature applications, low signal losses and wide frequency range of signal transmission enabling for better detection of materials inside the pipe/container. The nano materials can vary widely that would lead to implementation of wide range of impedance for single or multiple layer matching for signal improvement.

(64) In a similar design, FIG. 9A is a diagrammatical view illustration of a system for non-intrusive material identification 310, in accordance with exemplary embodiments of the present disclosure, which illustrates a transducer 320 transmitting signal nano technology enabled signals, electrically or mechanically induced signals. In particular, FIG. 9A depicts electrical or mechanical stimulation of ultrasound waves into oriented array of single- or multiwalled-nanotubes, where electrical or mechanical wave stimulating surface 330 or mechanism is placed on the top of the nanotubes layer. This nanotube oriented array couplant design may provide improved signal transmission and noise range filtering/selection as well as high-temperature applications, low signal losses and wide frequency range of signal transmission enabling for better detection of materials inside the pipe/container. The nano materials can vary widely that would lead to implementation of wide range of impedance for single or multiple layer matching for signal improvement.

(65) FIG. 9B is a diagrammatical view illustration of a system for non-intrusive material identification 310, in accordance with exemplary embodiments of the present disclosure. As shown, FIG. 8B illustrates electrical or mechanical wave stimulating surface 330 which are laced on the top of the nano-tube layer, where the graphene like two dimensional sheets can be oriented perpendicular or horizontal in relation to the surface of the wall of the container. These multiple

nanotube graphene like-sheets couplant designs may provide improved signal transmission and noise range filtering/selection as well as high-temperature applications, low signal losses and wide frequency range of signal transmission enabling for better detection of materials inside the pipe/container. The nano materials can vary widely that would lead to implementation of wide range of impedance for single or multiple layer matching for signal improvement.

(66) FIG. 10 illustrates a diagrammatical view illustration of a system for non-intrusive material identification 310, in accordance with exemplary embodiments of the present disclosure. In particular, FIG. 10 depicts how Multiwalled nanotubes can detect even the smallest signals since the innertubes are moving nearly without friction. As shown, a laser or optical sensor 340 is positioned to emit a signal 342 to a detector or optical sensor 344 which contacts the nanotube array 320. A second signal 346 may be used to determine a distance between the optical sensors 340, 344. The optical sensor 340 may emit the signal 342 as a laser sensing a coherent light beam to the surface of the nanotube array 320, while the signal 246 is for comparison of the optical sensors to determine motion remotely using reflected natural light, such as UV, visible, or IR light. The motion can be detected by sensor 344, which may be a laser interferometer, for instance. The pair of optical sensors can use this information to triangulate and determine the pattern of motions to be evaluated, thereby detecting even the smallest signals.

(67) The use of the enhanced nanotube couplant or gel may provide significant benefits in the industry of acoustic signal transmission. Carbon nanotubes or sphere-enhanced couplant can be understood using the average impedance for a class of materials. As such, the only viable way to increase the energy of the signal is to reduce the impedance of the transducer by adding a modified carbon nanotube layer on the faces of the transducer with the desired impedance suitable for each application or class of applications. This is depicted in Tables 5-7:

(68) TABLE-US-00005 TABLE 5 Steps 1 PZT Ceramic Nano Gel Optimal Reflection

Transmission	Transmission	28.50	8.35	29.92%	70.08%	70.08%	2	CNT PVC	9.67	18.94%	81.06%
56.81%	8.35	3.2844									

(69) TABLE-US-00006 TABLE 6 Steps 1 PZT Ceramic CNT Optimal Reflection Transmission

Transmission	Transmission	28.50	8.35	29.92%	70.08%	70.08%	2	CNT PE	7.15	8.35	1.794	41.75%	58.25%
40.82%													

(70) TABLE-US-00007 TABLE 7 Steps 1 PZT Ceramic CNT Optimal Reflection Transmission

Transmission	Transmission	28.50	8.35	29.92%	70.08%	70.08%	2	CNT PE	8.21	8.35	2.3674	31.14%	68.86%
48.26%													

(71) From Tables 5-7, it can be seen that the transducer impedance needs to be closer to the class of the material that has been identified. Thus, for air, it needs to be in the range of 600-800 Rayls, whereas for polymers, between 6-8 Mrayls, and for metals, approximately around 50 Mrayls.

(72) In some situations, a pipe formed with a polymer pipe wall, such as polyethylene (PE) can lead to a 50% signal increase, as shown in Table 8:

(73) TABLE-US-00008 TABLE 8 Steps PZT Ceramic CNT/Grphn-1 Optimal Reflection

Transmission	Transmission	28.50	16.80	6.67%	93.33%	93.33%	1	CNT/Grphn-1	16.5	16.24	16.80	
9.25	8.40%	91.60%	85.49%	2	CNT/Grphn-2	CNT/Grphn-3	8.20	9.25	4.00	84.30%	72.07%	3
CNT/Grphn-3	Stainless Long	4.08	4.00	1.8	14.39%	85.61%	61.70%					

Exponentially reducing layers may help increase the energy that will cross into the material. It is noted that the optimization may advantageously be done top down, layer by layer. When the top layer impedance produces a maximum transmission, then it may be possible to move to the bottom layer.

(74) FIG. 11 illustrates an example of the subject disclosure. For instance, in FIG. 11, there is potential for 50% signal energy improvement when using multilayered couplant for ultrasonic testing of materials. As shown in FIG. 11, the graph depicts a curve of diminishing return. Otherwise increased numbers of perfectly matched layers has potentially 3-5 layers as the optimal result. Detecting difficult materials may however warrant more layers. Adapting nanotubes like

CNTs by adding other materials than carbon may yield the desired results. The table in FIG. 11 illustrates the numerical values corresponding to the diminishing return.

(75) Exemplary values of the polymer pipe wall example for Polyethylene (PE) are depicted in Table 9:

(76) TABLE-US-00009 TABLE 9 Steps 1 PZT Ceramic Perfect Layer Optimal Reflection Transmission Transmission 28.50 7.16 35.81% 64.19% 64.19% 2 Perfect Layer PE 7.16 7.16 1.8 35.79% 64.21% 41.22%

(77) Better impedance may be achieved for layers adjusted for specific classes of pipe/container materials, such as, for instance, thermoplastic pipes. In particular, Tables 10-12 show the performance of using Ultrigel II couplant for each pipe wall. Pipes formed from PEX and also PE-RT can be combined with a thin aluminum middle layer to alter performance. In this situation, the aluminum may serve two purposes: as an oxygen barrier and to increase the longitudinal stiffness of the pipe. Details of this alteration are shown in Tables 10-12:

(78) TABLE-US-00010 TABLES 10-12 Steps 1 PZT Ceramic Ultrigel II Optimal Reflection Transmission Transmission 28.50 1.80 77.65% 22.35% 22.35% 2 Ultrigel II PVC 9.67 8.52% 91.48% 20.45% 1.80 3.2844 Steps 1 PZT Ceramic Ultrigel II Optimal Reflection Transmission Transmission 28.50 1.80 77.65% 22.35% 22.35% 2 Ultrigel II PE 7.15 1.80 1.794 0.00% 100.00% 22.35% Steps 1 PZT Ceramic Ultrigel II Optimal Reflection Transmission Transmission 28.50 1.80 77.65% 22.35% 22.35% 2 Ultrigel II PE 8.21 1.85% 98.15% 21.94% 1.80 2.3674

(79) Tables 13-18, below, illustrate using specific material adapted couplant rather than material class adapted couplant leads only to marginal improvement in signal energy.

(80) TABLE-US-00011 TABLE 13 Steps 1 PZT Ceramic Nano Gel Optimal Reflection Transmission Transmission 28.50 8.35 29.92% 70.08% 70.08% 2 CNT PVC 9.67 8.35 3.2844 18.94% 81.06% 56.81%

(81) TABLE-US-00012 TABLE 14 Steps 1 PZT Ceramic CNT Optimal Reflection Transmission Transmission 28.50 8.35 29.92% 70.08% 70.08% 2 CNT PE 7.15 8.35 1.794 41.75% 58.25% 40.82%

(82) TABLE-US-00013 TABLE 15 Steps 1 PZT Ceramic CNT Optimal Reflection Transmission Transmission 28.50 8.35 29.92% 70.08% 70.08% 2 CNT PE 8.21 8.35 2.3674 31.14% 68.86% 48.26%

(83) TABLE-US-00014 TABLE 16 Steps 1 PZT Ceramic Nano Gel Optimal Reflection Transmission Transmission 28.50 9.67 24.34% 75.66% 75.66% 2 CNT PVC 9.67 9.67 3.2844 24.30% 75.70% 57.28%

(84) TABLE-US-00015 TABLE 17 Steps 1 PZT Ceramic CNT Optimal Reflection Transmission Transmission 28.50 7.15 35.87% 64.13% 64.13% 2 CNT PE 7.15 7.15 1.794 35.86% 64.14% 41.14%

(85) TABLE-US-00016 TABLE 18 Steps 1 PZT Ceramic CNT Optimal Reflection Transmission Transmission 28.50 8.21 30.55% 69.45% 69.45% 2 CNT PE 8.21 8.21 2.3674 30.51% 69.49% 48.26%

(86) Together, these tables show that it is more practical to select one couplant design for class of pipes/containers. For instance, thermoplastic pipe class couplant performs reasonably well that the one optimized for each plastic, and it has been found that 3-5 layers are optimal due to diminishing return of the performance curve, as indicated in FIG. 11. Glass has very similar impedance to plastics, but the physical characteristics for the attachment of the sensor, and glass may require different adhesive properties than on the plastic pipes.

(87) It should be emphasized that the above-described embodiments of the present disclosure, particularly, any “preferred” embodiments, are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the disclosure. Many variations and modifications may be made to the above-described embodiment(s) of the disclosure without departing substantially from the spirit and principles of the disclosure. All such modifications and

variations are intended to be included herein within the scope of this disclosure and the present disclosure and protected by the following claims.

Claims

1. A system for improved, non-intrusive material identification, the system comprising: a vessel holding or transporting at least one quantity of a fluid; at least one nanotube transducer positioned on an exterior surface of a sidewall of the vessel, the nanotube transducer formed by a nanotube array positioned within a housing of the nanotube transducer, wherein the nanotube array within the nanotube transducer creates a vibration medium that substantially reduces acoustic signal degradation and improves a reflection of an acoustic signal emitted from the at least one nanotube transducer; and a computerized device in communication with the at least one nanotube transducer, the computerized device having a processor and a memory, wherein the computerized device determines a material identification of the quantity of fluid based on, at least in part, the reflection of the acoustic signal.
2. The system of claim 1, wherein the at least one nanotube transducer further comprises first and second nanotube transducers, wherein the vessel is positioned between the first and second nanotube transducers.
3. The system of claim 2, wherein the acoustic signal is transmitted between the first and second nanotube transducers.
4. The system of claim 1, wherein the nanotube array further comprises one of carbon nanotubes (CNT) or multi-walled carbon nanotubes (MWCNT).
5. The system of claim 1, further comprising at least one electrical or mechanical wave stimulating surface positioned on the top of the nanotube array of the at least one nanotube transducer.
6. The system of claim 1, wherein the at least one nanotube transducer comprises first and second nanotube transducers, the first and second nanotube transducers positioned on a single sidewall of the vessel.
7. The system of claim 1, wherein the nanotube array of the nanotube transducer further comprises a plurality of graphene-like sheets positioned substantially parallel to one another.
8. The system of claim 7, further comprising at least one electrical or mechanical wave stimulating surface positioned on the top of the plurality of graphene-like sheets.
9. A system for improved acoustic signal propagation through a medium, the system comprising: a vessel holding or transporting at least one quantity of a fluid; at least two acoustic transducers positioned on an exterior surface of a sidewall of the vessel; and at least one nanotube layer positioned on an exterior surface of the sidewall of the vessel and between each acoustic transducer of the at least two acoustic transducers, wherein an acoustic signal emitted from one of the at least two acoustic transducers is reflected between an interior sidewall of an interior surface of the vessel and an interior surface of the exterior sidewall where the nanotube layer is positioned, wherein the nanotube layer improves a reflection of the acoustic signal emitted from the one of the at least two acoustic transducers.
10. The system of claim 9, wherein the acoustic signal emitted by the one of the at least two acoustic transducers is received by a second of the at least two acoustic transducers.
11. The system of claim 9, wherein the nanotube layer further comprises at least one vertically aligned nanotube array.
12. The system of claim 11, further comprising at least one electrical or mechanical wave stimulating surface positioned on the top of the at least one vertically aligned nanotube array.
13. The system of claim 12, wherein the at least one vertically aligned nanotube array further comprises a plurality of graphene-like sheets positioned substantially parallel to one another.

14. The system of claim 13, further comprising at least one electrical or mechanical wave stimulating surface positioned on the top of the plurality of graphene-like sheets.
